

Landfill_emissions_r1.R

April 4, 2025

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Municipal_solid_waste	<i>Create gridded municipal solid waste methane emissions maps</i>
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Description

‘Municipal_solid_waste’ writes up to 4 netcdf files of gridded methane emissions from municipal landfills, as well as optional visuals and more easily read csv files

Usage

```
Municipal_solid_waste(  
  input_directory,  
  domain,  
  domain_template,  
  state_name_list,  
  output_directory,  
  inventory_year,  
  verbose,  
  GHGI_landfill_total,  
  ghgrp_facility_info,  
  landfill_ghgrp_reported,  
  landfill_ghgrp_modeled,  
  landfill_ghgrp_collection_efficiency,  
  plot_directory,  
  County_Tigerlines,  
  State_Tigerlines  
)
```

Arguments

- input_directory
Character providing the full filepath to save/load raw input data
- domain
SpatVector polygon outlining the desired output area

domain_template	SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system
state_name_list	Character vector listing all states within the desired domain
output_directory	Character providing the full filepath to save processed data
inventory_year	Numeric indicating the desired year of data to use.
verbose	Logical indicating whether to save additional output. This includes 2 csv files providing the downloaded information used for all landfills within the domain, separated between LMOP and GHGRP. It also includes up to 4 plots of the gridded methane emissions on log scales, one for LMOP facilities and one for each variation of the GHGRP facilities.
GHGI_landfill_total	Numeric. Pulled from config file. The total national emissions from municipal solid waste from the GHGI for the inventory_year in kilotons per year, equivalent to gigagrams per year. The GHGI is available at https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks and the value can be found in the table titled 'CH4 emissions from Landfills (kt)' as row 'MSW net CH4 Emissions'.
ghgrp_facility_info	Data.frame with the GHGRP location data for all years and states. See https://www.epa.gov/enviro/enr/data-service-api
landfill_ghgrp_reported	Logical. Pulled from config file. Whether or not to use reported GHGRP values.
landfill_ghgrp_modeled	Logical. Pulled from config file. Whether or not to overwrite reported GHGRP values with the modeled emissions (HH-6) for landfills with gas collection systems.
landfill_ghgrp_collection_efficiency	Logical. Pulled from config file. Whether or not to overwrite reported GHGRP values with the collection efficiency based emissions (HH-8) for landfills with gas collection systems.
plot_directory	Character providing the full filepath to save figures. Only relevant if verbose = TRUE.
County_Tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if verbose=TRUE.
State_Tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if verbose=TRUE.
LMOP_file	Character providing the full filepath to the landfill methane outreach program's landfill-level only data excel file available at https://www.epa.gov/lmop/landfill-technical-d . Only the LMOP_database tab is used and only the variables GHGRP_ID, latitude, and longitude are used. Columns represent separate variables and rows represent separate landfills. There is an example file in the package's datasets folder that has been successfully used in this code available for reference.

Details

This function calculates and grids methane emissions from municipal landfills. It uses the Environmental Protection Agency's (EPA) Greenhouse Gas Reporting Program (GHGRP) emissions when available. It then calculates the difference between national GHGRP emissions and the EPA national Greenhouse Gas Inventory (GHGI) emissions (GHGI - GHGRP) and distributes this residual equally to all facilities in the EPA Landfill Methane Outreach Program (LMOP) that are not already accounted for by the GHGRP.

The necessary GHGRP and LMOP data will be automatically downloaded.

Landfills have 2 options for reporting their emissions to the GHGRP - equation HH-6 and HH-8. HH-6 is based on a first order decay model, HH-8 is based on the collection efficiency of a gas collection system. Landfills can choose to treat either as the reported value, but both are provided to the GHGRP. If set in the config, the modeled values (HH-6) or collection efficiency (HH-8) can be used rather than the reported values when a gas collection system exists. Note the GHGRP total used for the LMOP calculation will be the reported values regardless, as these are what are used in the GHGI.

The GHGRP includes only facilities that emit at least 25,000 metric tons of carbon dioxide equivalent while the GHGI is intended to capture all national emissions and LMOP is a voluntary program with location and other details, but no emissions information. GHGRP data is available starting in 2010 and generally is about 2 years behind present day, the GHGI is available starting in 1990 and is updated approximately in sync with the GHGRP, and the LMOP is updated more frequently, sometimes multiple times per year. All three datasets are annual. The GHGRP and LMOP are at the facility scale while the GHGI is national totals.

The GHGI is available at <https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and->

The GHGRP is available at <https://www.epa.gov/ghgreporting> LMOP is available at <https://www.epa.gov/lmop/landfill-technical-data>

LMOP and GHGRP gridded emissions maps are saved separately.

Value

Nothing is returned from the function, but the main outputs are up to 4 netcdf files of the methane emissions of municipal landfills. They are titled "MSW_GHGRP_reported.nc", "MSW_GHGRP_modeled.nc", "MSW_GHGRP_collection_efficiency.nc", and "MSW_LMOP.nc".

If verbose is set to TRUE, then "MSW_GHGRP_all.csv" and "MSW_LMOP_all.csv" are also saved. The csvs include all downloaded variables for landfills within the domain.

Author(s)

Joe Pitt, <madeup@wisc.edu>

Kris Hajny, <blank@fake.edu>

Israel Lopez-Coto, <test@test.edu>

Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
             ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
             xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
             crs=grid_crs)
```

```

grid_polygon <- vect(ext(domain_template))
crs(grid_polygon) <- grid_crs

ghgrp_facility_info <- read.csv("~/../Desktop/GHGRP/facility_info.csv")
ghgrp_facility_info$county_fips <- sprintf("%05d",ghgrp_facility_info$county_fips)
ghgrp_facility_info$zip <- sprintf("%05d",ghgrp_facility_info$zip)

Municipal_solid_waste(domain=grid_polygon,
                      domain_template=grid,
                      state_name_list=c("DE", "MD", "NJ", "NY", "PA"),
                      output_directory=~../Desktop/out/",
                      input_directory=~../Desktop/in/",
                      inventory_year=2018,
                      verbose=TRUE,
                      GHGI_landfill_total = 3943,
                      ghgrp_facility_info=~../Desktop/in/GHGRP/facility_info.csv",
                      landfill_ghgrp_reported=TRUE,
                      landfill_ghgrp_modeled=TRUE,
                      landfill_ghgrp_collection_efficiency=TRUE,
                      plot_directory=~../Desktop/plots/"
                      County_Tigerlines=vect("~/../Desktop/in/County_Tigerlines/tl_2018_us_county.shp"),
                      State_Tigerlines=vect("~/../Desktop/in/State_Tigerlines/tl_2018_us_state.shp"))

```

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